

Tinsae Melkamu

Backend Developer (Node.js / Express) TypeScript | PostgreSQL | MongoDB | REST APIs

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Profile

Backend-focused software engineer specializing in building scalable RESTful APIs using Node.js, Express, and TypeScript. I build secure, production-style backend systems with strong focus on authentication, database design, and clean architecture.

I have hands-on experience developing real-world backend systems including AI-integrated platforms, authentication APIs, and financial tracking systems using PostgreSQL and MongoDB. My focus is on building reliable, maintainable, and well-structured backend services.

Core strengths include API development, authentication (JWT), database modeling, middleware design, and system reliability. I emphasize clean code, debugging, and production-ready backend architecture.

EXPERIENCE & PROJECTS

Student Expense Tracker API — Personal Finance Management System 🔗,

Node.js, Express, TypeScript, PostgreSQL, Prisma, JWT, Zod, Vitest, Supertest

- Built a secure backend API for tracking income, expenses, and categories with authentication and user-specific data isolation.
- Designed a relational PostgreSQL schema using Prisma with optimized data modeling and indexed queries.
- Implemented JWT authentication, filtering, monthly analytics, and backend security practices including input validation, rate limiting, and IDOR prevention.
- Wrote unit tests using Vitest and integration tests using Supertest to validate API behavior and improve backend reliability.

DevArcAI — AI-Powered Developer Documentation Assistant 🔗,

Node.js, Express, ChromaDB, SQLite, Gemini API, RAG-based retrieval system

- Built an AI-powered backend system using RAG-based retrieval to summarize technical documentation via vector search and LLM integration.
- Designed REST APIs for document ingestion, chat workflows, and admin management.
- Implemented semantic search with ChromaDB and modular backend architecture for scalability.

Secure Image Upload API — Authentication & Media Management System 🔗,

Node.js, Express, MongoDB, Cloudinary, Multer, JWT

- Built a secure authentication and image management API with upload, retrieval, and deletion features.
- Implemented JWT-based authentication with role-based access control and protected routes.
- Integrated Cloudinary and Multer for scalable file storage and handling.

Education

BSc in Computer Science and Engineering, Adama Science and Technology University

2023 – 2028

Relevant Coursework:

- Data Structures & Algorithms
- Database Management Systems (SQL, relational modeling)
- Software Engineering (SDLC, documentation practices)

Technical Skills

Languages

- JavaScript(ES6+)
- TypeScript

Databases & Data modeling

- PostgreSQL
- MongoDB
- Prisma ORM
- SQLite

Back-end Development

- Node.js, Express.js
- RESTful API design & architecture
- Authentication and Authorization (JWT, cookie-based sessions)
- Centralized error handling & Middleware design.
- Role-based access control(RBAC)
- Secure backend system design

Tools

- Git & Github
- Postman
- Environment-based configuration
- Deployment(Render)